

IC-21: CALORIMETRY

Rev 5-09-2023

21.1 OBJECTIVE

The purpose of this experiment is to measure the specific heat of metals, and the latent heat of fusion of ice, through heat balance.

21.2 EQUIPMENT

1. Calorimeter
2. Thermometer / temperature sensor
3. Boiler / tripod
4. Sample metal objects
5. Triple-beam balance
6. Water
7. Ice
8. String
9. Paper towel



21.3 THEORY

The measurement of heat by the method of mixtures makes use of the principle that when heat interchange takes place between two bodies initially at different temperatures, the quantity of heat lost by the hotter body is equal to that gained by the cooler body, and they both come to some final, equilibrium temperature. In this it is assumed that no heat is lost to, or gained from, the surroundings.

Specific heat is defined as the quantity of heat (in calories or joules) that is required to raise the temperature of unit mass of a substance by one degree. In SI units, the units of specific heat are J/kg.C. In the experimental determination of specific heat, we shall mix a metal object at a relatively higher temperature to water which is at a lower temperature. The final temperature of the mixture (also called the equilibrium temperature) will be measured, as well as the initial temperatures of the metal and water. Since the material of the calorimeter itself is also going to absorb some heat, it also has to be included in the calculations. Then the specific heat, c , is given by:

$$|\text{Heat gained by water and calorimeter}| - |\text{Heat lost by hot metal}| = 0$$

Which yields:

$$|M_w c_w (T_e - T_{iw}) + M_c c_c (T_e - T_{ic})| - M_m c_m (T_3 - T_e) = 0$$

$$M_w c_w (T_e - T_{iw}) + M_c c_c (T_e - T_{ic}) + M_m c_m (T_e - T_3) = 0 \quad (1)$$

$$(M_w c_w + M_c c_c)(T_e - T_{iw}) + M_m c_m (T_e - T_3) = 0 \quad (2)$$

Where:

' M ' is mass and ' c ' is specific heat, and their subscripts ' w ', ' c ' and ' m ' refer to water, calorimeter and metal respectively.

T_{iw} , T_{ic} and T_{im} are the initial temperatures of water, calorimeter and metal, respectively.

T_e is the final equilibrium temperature of the calorimeter including all three materials.

The Latent heat is the amount of heat required to change unit mass of a substance from solid state to liquid state, or from a liquid state to gaseous state, without a change in temperature. The latent heat of fusion is the latent heat when a solid turns to liquid. Its SI units are J/kg. In the experiment, we shall measure the temperature of the calorimeter plus water before and after mixing it with ice. The masses and temperatures of the water and ice before and after mixing will be measured. Then, the latent heat will be found from:

Heat lost by water and calorimeter = Heat gained by ice to warm up to 0°C + Heat gained by ice at 0°C to melt into water at 0°C + heat gained by melted ice (water) to come to T_e .

$$(M_w c_w + M_c c_c)(T_e - T_{iw}) + M_{ice} c_{ice}(0 - T_{i-ice}) + M_{ice} L_{ice} + M_{ice} c_w(T_e - 0) = 0 \quad (3)$$

Where the subscript *ice* refers to ice, T_{i-ice} in initial temperature of ice, and L_{ice} is latent heat of fusion of ice.

21.4 EXPERIMENTAL PROCEDURE

Specific heat

1. Weigh the empty calorimeter (inner cup only, M_c).
2. Pour cold water in the calorimeter cup till it is about half full, and weigh it, and replace it in the insulating jacket. The temperature of this water should be about as many degrees lower (about 3°C) than the room temperature water, as the final equilibrium temperature is expected to be above it. This will reduce the errors due to radiation heat loss.
3. Record the temperature of the water (T_{iw}).
4. Fill the boiler with enough water to cover the metal cylinder plus a few inches to spare. Place it on the heater. Turn on the heater, and let the water come to a boil.
5. Weigh the metal object (M_m). Lower it in the boiling water by means of a thread. Make sure it is completely immersed, but not touching the sides or bottom of the boiler. Keep it in there till it comes to the temperature of the hot water (T_{mi}).
6. Quickly but carefully transfer the metal object from the boiler to the calorimeter. Stir the water and record the equilibrium temperature.
7. Measure and record the temperature of the boiling water.

Latent Heat

8. Repeat procedures 1, 2 and 3. The cold water should be about 10°C above room temperature (since we expect the final equilibrium temperature to be about 10°C below room temperature).
9. Dry some pieces of ice on a paper towel and add them to the water in the calorimeter. Avoid touching the ice (to prevent melting).
10. Stir the mixture. Record the equilibrium temperature (when ice is completely melted).
11. Weigh the calorimeter with water and melted ice.

NOTE: The thermometer takes the temperature of the water that it is dipped in, but after taking it out, it cools down in air quite slowly.

21.5 CALCULATIONS

1. Calculate the specific heat from equation 1
2. Calculate the latent heat from equation 2
3. Calculate the percent errors.

$$\text{Percent difference} = 100 \% \times [\text{Actual} - \text{measured value}] / \text{actual value}$$

21.6 PRECAUTIONS

1. The boiler is hot. Use with care.
2. Do not fill the boiler with too much water, or it will take too long for the water to reach a high temperature.
3. Do not drop the metal in the boiler, especially if you are using the beaker. Ease it in slowly.
4. Handle the metal with a thread tied to it.
5. Some thermometers have a delay (take time to reach equilibrium).
6. Use the larger objects available to you, so that the temperature changes are larger.
7. Dry the ice on a paper towel before placing it in the calorimeter.

Date: _____

Material of Calorimeter: _____

Specific heat of calorimeter: _____

Measurements of Specific Heat:**Object Labelled:** _____

Mass of empty Calorimeter (M_c)	
Mass of calorimeter + water	
Mass of water (M_w)	
Mass of metal (M_m)	
Initial temperature of cold water (T_{wi})	
Temperature of boiling water and metal (T_{mi})	
Final equilibrium temperature (T_e)	

Calculated value of specific heat _____

Apparent nature of metal: _____

Book value of specific heat:

Percent error: _____

Object Labelled: _____

Mass of empty Calorimeter (M_c)	
Mass of calorimeter + water	
Mass of water (M_w)	
Mass of metal (M_m)	
Initial temperature of cold water (T_{wi})	
Temperature of boiling water and metal (T_{mi})	
Final equilibrium temperature (T_e)	

Calculated value of specific heat _____

Apparent nature of metal: _____

Book value of specific heat:

Percent error: _____

Measurement of Latent Heat of Ice

Mass of empty Calorimeter (M_c)	
Mass of calorimeter + water	
Mass of water (M_w)	
Initial temperature of water (T_{wi})	
Initial temperature of ice (T_{i-ice})	
Final equilibrium temperature (T_e)	
Mass of calorimeter + water + melted ice	
Mass of ice	

Calculated value of Latent Heat: _____ Book value of Latent Heat of ice:

Percent error: _____

21.8 REPORT SUBMISSION

Upload the following in the Report for this Lab:

		Points in report
1.	The completely filled up "Report Form" for Specific Heat for first solid Units missing	10
2.	The completely filled up "Report Form" for Specific Heat for second solid. Units missing	10
3.	The completely filled up "Report Form" for Latent Heat of Ice. Units missing	10
4.	Attach pictures of the two objects used for the Specific Heat	6
5.	Sample Calculations Calculation error	5
6.	Sources of Error in this experiment. Indicate the major sources of error. Human Error, Calculation Error, and Rounding Error.	5
7.	Your Conclusions from this experiment	10
	Total	55

21.9 ADDITIONAL INFORMATION

1. Values of some Specific Heats are:

A-Zinc 0.39 J/gC B-Aluminum 0.90 J/gC C-Stainless Steel 0.45 J/gC
D-Copper 0.39 J/gC E-Brass 0.35 J/gC

2. You can see the entire experiment being performed (at home due to covid) at:
https://youtu.be/Nd_xEKgkUMo

3. Hewitt Drew It No. 79. Energy of Phase Changes
http://www.hewittdrewit.com/Links/79_Energy_of_Phase_Changes%C2%A0.html

4. Hewitt Drew It 80. More on Phase Changes:
http://www.hewittdrewit.com/Links/80_More_on_Phase_Changes%C2%A0.html

21.10 SAMPLE DATA

Material of Calorimeter: Aluminum

Specific heat of calorimeter: 900 J/kg°C

Specific Heat of Object Labelled: _____

Mass of empty Calorimeter (M_c)	58 g
Mass of calorimeter + water	156 g
Mass of water (M_w)	98 g
Mass of metal (M_m)	66 g
Initial temperature of cold water (T_{wi})	25.0 °C
Temperature of boiling water and metal (T_{mi})	100 °C
Final equilibrium temperature (T_e)	30.0 °C

Calculated value of specific heat _440 J/kgC_

Apparent nature of metal: _Brass_

Book value of specific heat: _401 J/kg°C_

Percent error: _9.7 %_

Latent Heat of Ice

Mass of empty Calorimeter (M_c)	56.0 g
Mass of calorimeter + water	136.0 g
Mass of water (M_w)	80.0 g
Initial temperature of water (T_{wi})	27.0 °C
Initial temperature of ice (T_{i-ice})	0.0 °C
Final equilibrium temperature (T_e)	11.6°C
Mass of calorimeter + water + melted ice	154.0 g
Mass of ice	18.0 g

Calculated value of Latent Heat: _281 J/g_

Book value of Latent Heat of ice: 334 J/kg

Percent error: _16%_